

Aspen Johnson

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SUMMARY

Data Science student and research assistant with experience in data curation, AI response evaluation, model validation, technical documentation, and applied research across econometrics and physics. Skilled at cleaning and structuring datasets, reviewing model outputs, documenting inconsistencies, and communicating technical findings clearly. Interested in AI evaluation, data quality, physical AI, and computational modeling for real-world systems.

ROLE-RELEVANT QUALIFICATIONS

- Valid Florida driver's license with 4 years of U.S. driving experience.
- Practical understanding of U.S. traffic laws, road signs, lane behavior, right-of-way rules, speed limits, traffic signals, and pedestrian safety.
- Experience evaluating AI-generated responses, model outputs, and technical explanations for accuracy, coherence, and clarity.
- Strong visual reasoning, spatial judgment, pattern recognition, and evidence-based decision-making developed through driving experience, athletics, physics coursework, and applied research.

EDUCATION

Valdosta State University

Incoming B.S. in Data Science: Computational Science & Engineering Track; Minors: Physics and Mathematics

Palm Beach State College

A.A. in STEM: Mechanical & Aerospace Engineering Track

June 2025 – May 2026, GPA: 3.20

PUBLICATIONS

[An empirical application of improved gradient scaling for score-driven volatility filters](#)

- *Applied Economics Letters*, Taylor & Francis (2025) | Co-authored with [Szabolcs Blazsek](#) (Mercer University) and [Adam Kobor](#) (New York University Investment Office).
- Contributed to data curation, analysis, software implementation, validation, visualization, and manuscript preparation for a peer-reviewed empirical modeling study.
- Selected as 1 of 15 research papers to present a lecture at the 2026 Atlanta Research Conference.

EXPERIENCE

Physics Undergraduate Researcher

Mar 2026 – Present

Valdosta State University

On-Site/Remote

- Collaborate with Professor S. Mubin on modeling stock market illiquidity as a bridge toward larger particle simulation research.
- Use Python-based tools including NumPy, Matplotlib, and Google Colab to analyze data, review visualizations, and document technical findings clearly.
- Review model outputs, visualizations, and research notes for accuracy, clarity, and consistency before faculty discussion.
- Evaluate AI-assisted summaries, model outputs, and visual evidence against source material to identify errors, inconsistencies, and unclear reasoning.

Econometrics Researcher

Jan 2025 – Dec 2025

Mercer University

On-Site/Remote

- Sourced and structured financial data from the Bloomberg Terminal, computing return series and descriptive statistics across candidate assets in Excel and R.
- Benchmarked 6+ volatility models (gjrgARCH, sGARCH, Beta-t-EGARCH) in R and GAUSS, selecting optimal asset universe based on data sufficiency and model fit.

- Debugged errors across R and GAUSS environments and documented line-level issues to support reproducibility, accuracy, and faculty review.
- Validated model outputs and documented errors, inconsistencies, and performance differences across competing volatility models for weekly faculty review.
- Co-authored peer-reviewed manuscript in $\text{\LaTeX}$ published in *Applied Economics Letters* (Taylor & Francis, 2025).

Financial Analyst Intern

May 2025 – Dec 2025

Zinzino USA

Remote

- Reduced chargeback processing time by 87% through workflow digitization and documentation streamlining.
- Refined the chargeback process within the first few weeks and created a clear reference guide to support onboarding, consistency, and cross-functional accuracy.
- Created clear process documentation that improved cross-functional understanding, consistency, and accuracy.
- Reviewed finance-related documentation and case details for completeness, accuracy, and clarity before sharing guidance with internal stakeholders.

PROJECTS

Econometric Modeling in Complex Physical Systems | In Progress

- Investigating the applicability of GARCH-family models to complex physical systems exhibiting variance clustering, including geomagnetic storms, seismic activity, and solar wind turbulence.
- Sourcing and structuring large-scale geophysical time series data from USGS, NOAA, and NCAR databases for empirical volatility modeling.
- Reviewing time-series patterns, anomalies, and model behavior to assess whether financial econometric methods can generalize to physical systems.
- Consulting seismic research labs to learn domain-specific considerations for interpreting earthquake time series and applying volatility-based modeling methods.

DIY Cyberdeck | In Progress | *Fusion 360, KiCad, PCB Design, Embedded Systems*

- Designing a portable computing device with attention to component layout, physical constraints, spatial reasoning, and technical documentation.

LEADERSHIP, COMMUNITY ENGAGEMENT & EXTRACURRICULARS

Division I Women's College Basketball Player | *Mercer University (2022 - 2025)*

Division II Women's College Basketball Player | *Valdosta State University (Present)*

Content Creation | *Created educational livestream content focused on problem sets, coding projects, and technical self-study.*

TECHNICAL SKILLS

AI/Data Review: AI response evaluation, LLM prompt testing, data annotation, data labeling, data quality review, visual evidence

Programming and Analysis: Python, R, GAUSS, Java, C/C++, JavaScript, Octave, Excel

Data Science: Data cleaning, exploratory data analysis, time-series analysis, model validation, statistical modeling, data visualization

Libraries & Tools: NumPy, Matplotlib, Seaborn, Google Colab, $\text{\LaTeX}$, Git, CLI, VS Code

Driving/Spatial Reasoning: Valid Florida driver's license, U.S. traffic laws, road-sign recognition, lane behavior, right-of-way